

Sri Lanka Institute of Information Technology



Faculty of Computing

IT2011 - Artificial Intelligence and Machine Learning

Lab 05 - Introduction to Generative AI

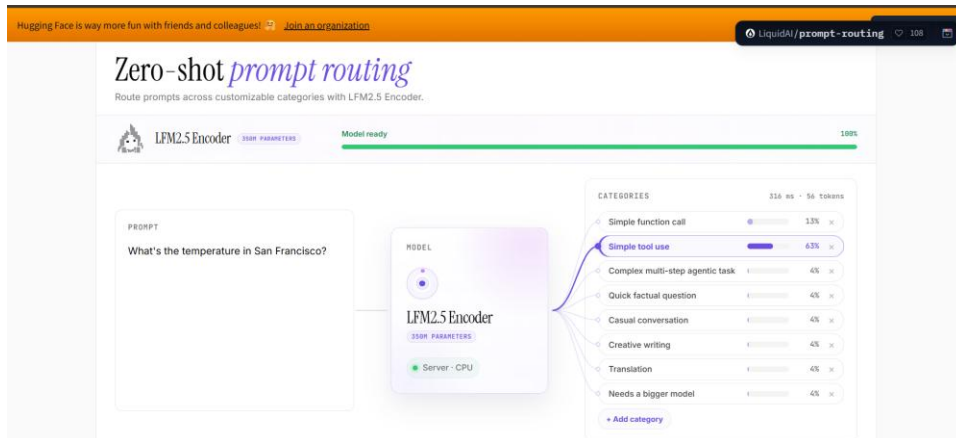
Submission: Exercises 2 to 6

IT25101663 - Faleel M. H.

Exercise 5: Hugging Face & Colab

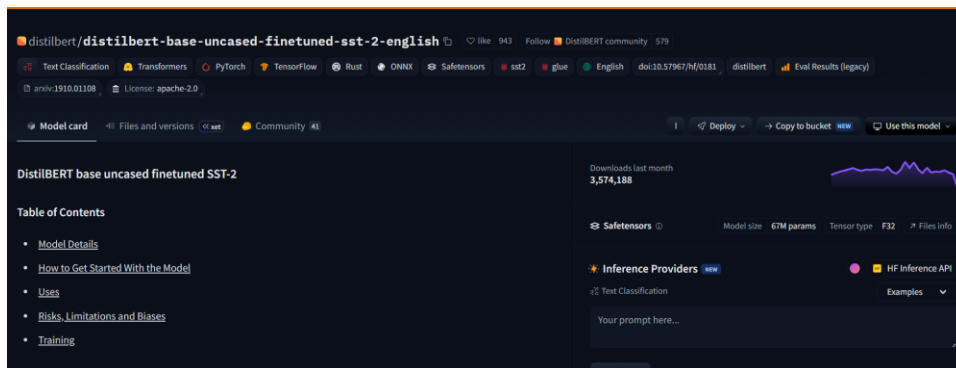
Spaces demo

Tried LiquidAI's "Zero-shot Prompt Routing" Space (Text Analysis category). Input: "What's the temperature in San Francisco?" → classified as 63% "Simple tool use."



Model details

distilbert-base-uncased-finetuned-sst-2-english - 67M params, 3.5M+ downloads/month, Apache-2.0, fine-tuned on SST-2 for sentiment classification. (Separate model from the one chosen for the Colab task below.)



Colab: summarization models

Note: pipeline("summarization") errored in this Colab environment (KeyError, task not registered). Worked around it using AutoModelForSeq2SeqLM directly.

Default model (distilbart-cnn-12-6): "Hugging Face is a pioneering open-source AI platform that empowers developers, researchers"

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Ple
config.json: 100% ██████████ 1.80k/1.80k [00:00<00:00, 31.2kB/s]
tokenizer_config.json: 100% ██████████ 26.0/26.0 [00:00<00:00, 465B/s]
vocab.json: 100% ██████████ 899k/899k [00:00<00:00, 9.13MB/s]
merges.txt: 100% ██████████ 456k/456k [00:00<00:00, 6.87MB/s]
pytorch_model.bin: reconstructing file: 100% ██████████ 1.22GB / 1.22GB, 69.2MB/s
pytorch_model.bin: downloading bytes: ██████████ 933MB, 45.9MB/s
model.safetensors: reconstructing file: 100% ██████████ 1.22GB / 1.22GB, 75.7MB/s
model.safetensors: downloading bytes: ██████████ 932MB, 48.7MB/s
[transformers] Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights: 100% ██████████ 358/358 [00:00<00:00, 2794.44it/s]
Hugging Face is a pioneering open-source AI platform that empowers developers, researchers
```

kobart-summary-v3 (Korean model, on English text): "Hugging Face is a pioneering op" - essentially a truncated echo, not a real summary, since the model isn't trained for English.

```
special_tokens_map.json: 100% ██████████ 692/692 [00:00<00:00, 16.7kB/s]
model.safetensors: reconstructing file: 100% ██████████ 496MB / 496MB, 43.5MB/s
model.safetensors: downloading bytes: ██████████ 468MB, 38.4MB/s
Loading weights: 100% ██████████ 260/260 [00:00<00:00, 3233.23it/s]
generation_config.json: 100% ██████████ 191/191 [00:00<00:00, 5.46kB/s]
[transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer TokenizersBackend. The clear
Hugging Face is a pioneering op
```

Own choice model: English -> French Translation

Model: Helsinki-NLP/opus-mt-en-fr (distinct from the sentiment model used for the model-details lookup above).

```
config.json: 100% ██████████ 1.42k/1.42k [00:00<00:00, 57.6kB/s]
tokenizer_config.json: 100% ██████████ 42.0/42.0 [00:00<00:00, 2.29kB/s]
source.spm: 100% ██████████ 778k/778k [00:00<00:00, 26.8MB/s]
target.spm: 100% ██████████ 802k/802k [00:00<00:00, 24.4MB/s]
vocab.json: 100% ██████████ 1.34M/1.34M [00:00<00:00, 28.3MB/s]
/usr/local/lib/python3.13/dist-packages/transformers/models/ Marian/tokenization_marian.py:176: UserWarning: Recommended: pip install sacremoses.
warnings.warn("Recommended: pip install sacremoses.")
pytorch_model.bin: reconstructing file: 100% ██████████ 301MB / 301MB, 26.4MB/s
pytorch_model.bin: downloading bytes: ██████████ 281MB, 25.0MB/s
model.safetensors: reconstructing file: 100% ██████████ 301MB / 301MB, 27.1MB/s
model.safetensors: downloading bytes: ██████████ 280MB, 24.0MB/s
Loading weights: 100% ██████████ 258/258 [00:00<00:00, 1333.75it/s]
generation_config.json: 100% ██████████ 293/293 [00:00<00:00, 5.63kB/s]
Hugging Face makes machine learning accessible to everyone.
→ Hugging Face rend l'apprentissage automatique accessible à tout le monde.
I would like to book a table for two tonight.
→ J'aimerais réserver une table pour deux ce soir.
This lab was more difficult than I expected.
→ Ce labo était plus difficile que je ne m'y attendais.
```

All three test sentences were translated into grammatically correct, natural French, confirming the model performs well within its trained language pair.